

Supplementary material for Prior-free relative 6D pose estimation of multiple object instances

1. Overview

This supplementary material provides additional analyses, quantitative results, and qualitative visualizations that complement the experiments reported in the main paper. In Sec. 2, we provide detailed statistics of the PRENCH benchmark, including the number of images, object instances, anchor-instance pairs, and objects in each dataset, as well as the distribution of instances per view and anchor-instance overlap. In Sec. 3, we present extended ablation studies that further analyze the behavior of PROSE under different levels of anchor-instance overlap. We then investigate the effect of cycle consistency refinement on each scene. These experiments provide a more detailed view of the impact of global correspondence refinement via cycle-consistency in the accuracy of predicted poses. In Sec. 4, we provide additional quantitative results for the correspondence refinement stage, including both per-scene and dataset-level comparisons before and after refinement. These results complement the evaluation in the main paper and further validate the effect of multi-instance reasoning across IC-BIN [1], IC-MI [5], and XYZ-IBD [3]. Finally, in Sec. 5, we present additional qualitative results illustrating the behavior of the semantic (DINOv2 [4]), geometric (dGeDi [2]), and fused feature representations. We visualize both the learned feature representations and the resulting inlier correspondences across textured everyday objects and textureless industrial objects. We additionally visualize individual cycle-consistency refinement examples, including both successful and failure cases, to clarify how cycle-consistency refinement modifies the initial pairwise correspondences.

2. Benchmark statistics

Fig. 1 and Tab. 1 summarize PRENCH. In total the benchmark comprises 1,166 images and 21,833 object instances across three datasets and 21 distinct objects, giving 20,667 anchor-instance pairs for evaluation. The datasets cover different number of per-view instances: XYZ-IBD averages 26.5 instances per view (up to 59), IC-BIN 9.8, and IC-MI only 2.7 (Fig. 1, top). This diversity is central to our analysis, as the cross-instance refinement exploits redundant correspondences; consistent with this, its benefit

is largest on the dense XYZ-IBD scenes and negligible on the sparse IC-MI ones (cf. Fig. 3 and the per-scene analysis in Tab. 2). The datasets also differ in difficulty, quantified by the anchor-instance overlap, splitting instances into easy ($\text{IoU} > 0.6$), medium ($0.3 < \text{IoU} < 0.6$), and hard ($0 < \text{IoU} < 0.3$) (Fig. 1, bottom). IC-BIN mainly contains low-overlap (hard) pairs., IC-MI contains medium range overlap, and XYZ-IBD is spread across the full range. Combined with a diverse object set (everyday textured objects in IC-BIN and IC-MI versus textureless, industrial parts in XYZ-IBD), PRENCH forms a broad evaluation setting that emphasizes relative pose estimation under both varying instance counts and varying degrees of overlap and occlusion.

3. Extended ablation studies

Figure 2 decomposes average recall by anchor-instance overlap, splitting instances into easy ($\text{IoU} > 0.6$), medium ($0.3 < \text{IoU} < 0.6$), and hard ($0 < \text{IoU} < 0.3$). As expected, accuracy decreases with overlap for every method, since a smaller visible surface overlap yields fewer reliable correspondences. PROSE attains the highest AR in almost every bin across the three datasets, and its margin is largest in the easy and medium regimes, *e.g.*, on XYZ-IBD it reaches 72.9%/48.8% (easy/medium) against 47.0%/25.0% for ConceptPose and 23.0%/10.0% for One2Any. The two baselines show no consistent ordering: ConceptPose is generally stronger on easy instances, while One2Any is more competitive on the hardest bins of IC-BIN and IC-MI. The only setting where PROSE is marginally surpassed is the hard split of IC-MI (37.8% vs. One2Any’s 42.0%), which comprises just 42 low-overlap instances. Overall, PROSE’s gains are not limited to easy cases but persist across the full difficulty spectrum, confirming its benefits in low-overlap scenarios.

Figure 3 reports the change in average recall produced by the refinement, $\Delta \text{AR} = \text{AR}_{\text{after}} - \text{AR}_{\text{before}}$, for every scene of the three datasets (green: improved; red: worsened; dashed line: dataset mean). The effect is strongly dataset-dependent and tracks the number of co-visible instances. On XYZ-IBD, whose bins contain many instances of the same industrial part, refinement improves 12 of 15 scenes for a mean gain of +2.4 AR, reaching up to +7 on the densest scenes. On IC-BIN the effect is essentially neutral (mean

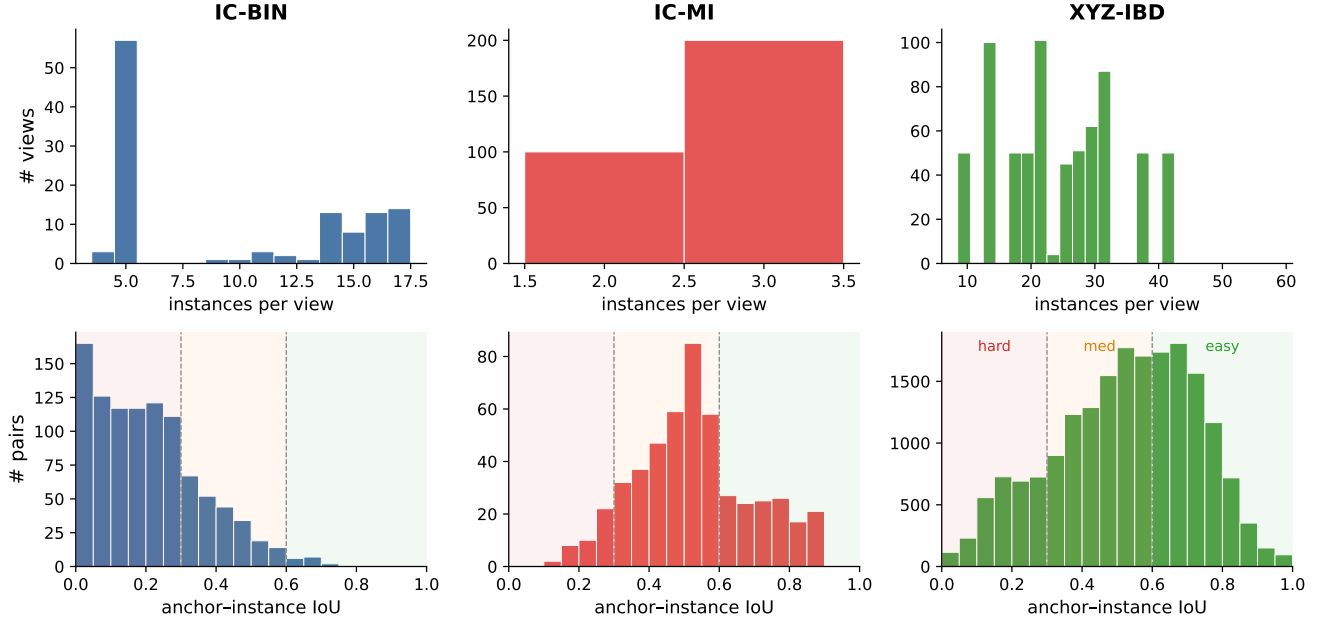


Figure 1. Top row: number of instances per view for each dataset. Bottom row: distribution of anchor–instance IoU, with the easy/medium/hard difficulty bands shaded (thresholds at 0.3 and 0.6). XYZ-IBD and IC-BIN contain many instances per image (mean 26.5 and 9.8, up to 59 and 17), whereas IC-MI has only ~ 3 ; correspondingly, IC-BIN is skewed toward low overlap (hard), IC-MI concentrates in the medium range, and XYZ-IBD spreads across all difficulties.

Table 1. PRENCH statistics across individual datasets (rows 1–3) and overall (row 4). Columns show, from left to right, the number of objects, the number of images, the number of object instances (with details about per-image average and maximum), and the number of anchor–instance pairs (with details about difficulty levels: easy for anchor–instance pairs with $\text{IoU} > 0.6$, medium with $0.3 \leq \text{IoU} \leq 0.6$, and hard with $\text{IoU} < 0.3$). *Two objects recur across IC-BIN and IC-MI, so the total number of distinct objects is 21 instead of 23.

	Dataset	#Obj	#Img	#Instances			#Pairs			
				Total	Avg	Max	Total	Easy	Medium	Hard
1	IC-BIN	2	116	1,134	9.8	17	1,018	15	230	757
2	IC-MI	6	300	800	2.7	3	500	140	318	42
3	XYZ-IBD	15	750	19,899	26.5	59	19,149	7,592	8,445	3,047
4	PRENCH	21*	1,166	21,833	13.0	59	20,667	7,747	8,993	3,846

+0.4), whereas on IC-MI, whose scenes contain only about three instances on average, refinement is marginally detrimental (mean -1.0), since too few overlapping cycles are available for the synchronization to exploit. This confirms the trend in Fig. 5 of the main paper: enforcing cycle consistency becomes beneficial when enough mutual references co-occur, and its gain grows with the number of instances. Crucially, the many-instance regime is exactly the setting our method targets (bin-picking of numerous identical objects, as in XYZ-IBD), so the refinement contributes its largest improvements precisely where the task demands them.

4. Additional quantitative results

Table 2 isolates the contribution of the refinement: for each scene we report the four metrics obtained when RANSAC is fed the initial nearest-neighbour correspondences (*bef.*) versus the refined ones (*aft.*), holding all other components fixed. On XYZ-IBD, refinement improves AR on 12 of the 15 scenes and reduces both rotation and translation error on almost all of them; the largest AR gains are +7.3 (scene 000005, $51.8^\circ \rightarrow 43.2^\circ$ RE, $23.0 \rightarrow 15.8$ mm TE), +6.2 (000060, $24.1^\circ \rightarrow 15.7^\circ$ RE) and +4.6 (000040, $19.3^\circ \rightarrow 12.0^\circ$ RE). On the remaining three scenes (000035, 000045, 000065) AR changes by at most 0.8 points, *i.e.* refinement leaves them essentially unchanged. On IC-BIN the effect is small but positive (AR and RE improve on both scenes),

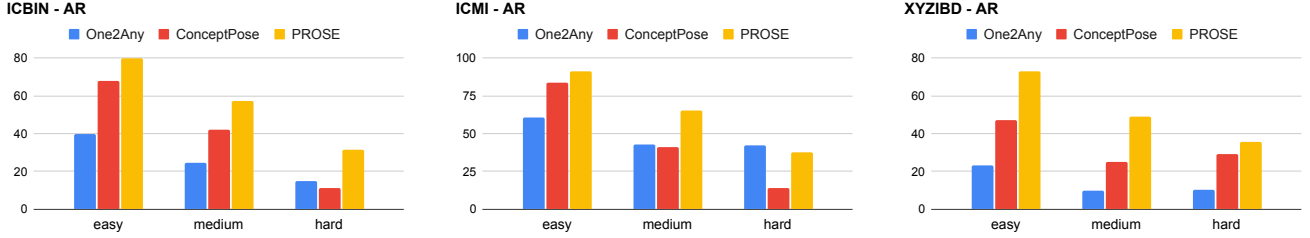


Figure 2. Per-difficulty average recall of PROSE vs. the One2Any and ConceptPose baselines on IC-BIN, IC-MI, and XYZ-IBD, with instances split by anchor–instance overlap into easy ($\text{IoU} > 0.6$), medium ($0.3 < \text{IoU} < 0.6$), and hard ($0 < \text{IoU} < 0.3$).

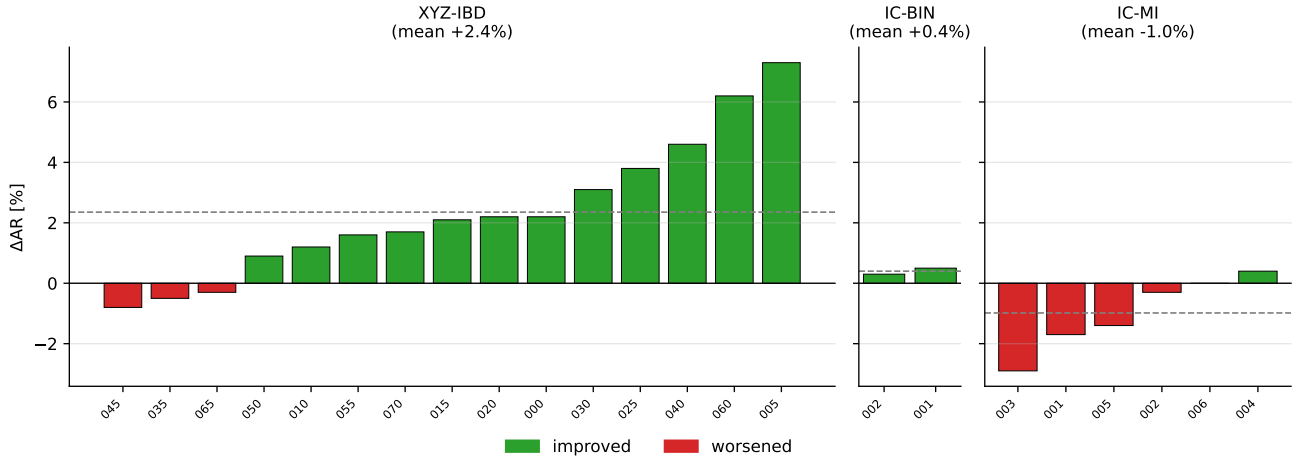


Figure 3. Change in average recall induced by the refinement, $\Delta\text{AR} = \text{AR}_{\text{after}} - \text{AR}_{\text{before}}$, for every scene of XYZ-IBD, IC-BIN, and IC-MI (bars sorted within each dataset; green = improved, red = worsened; dashed line = dataset mean). Refinement helps most on the many-instance XYZ-IBD scenes (mean +2.4), is neutral on IC-BIN (+0.4), and is slightly detrimental on the few-instance IC-MI (−1.0).

while on IC-MI it is neutral to slightly negative: ADD-S is already saturated at 100% and the other metrics move by at most a few points (largest drop −2.9 AR on 000003). This dataset-level pattern is consistent with the per-scene ΔAR summary of Fig. 3 and the instance-count trend of Fig. 5: refinement helps most where many instances co-occur (XYZ-IBD) and little where only a few are available (IC-MI, ~ 3 per scene). Overall, refinement never causes a large regression (worst per-scene AR drop 2.9 points) while yielding consistent gains in the many-instance regime that motivates our setting.

Table 3 reports the dataset-level effect of cycle-consistency refinement comparing the metrics obtained when RANSAC is fed the initial nearest-neighbour correspondences (*before*) versus the refined ones (*after*), with all other components held fixed. On XYZ-IBD refinement improves *every* metric: AR from 53.2 to 55.8 (+2.6 points), ADD-S from 90.6 to 92.1, rotation error from 32.1° to 28.6° (−3.5°), and translation error from 10.6 to 9.1 mm (−1.5 mm), a substantial and consistent gain. On IC-BIN the

effect is only marginal (AR +0.4, ADD-S +0.9, RE −0.3°), with translation error slightly worse (+0.5 mm). On IC-MI refinement is marginally detrimental (AR −0.9, RE +0.1°, TE +0.2 mm), while ADD-S is already saturated at 100%. This ordering (clear gains on XYZ-IBD, near-neutral on IC-BIN, slightly negative on IC-MI) matches the per-scene breakdown (Table 2, Fig. 3) and follows the instance-count trend of Fig. 5: synchronization exploits redundancy across co-visible instances, so its benefit scales with the number of instances per view (26.5 on XYZ-IBD versus 2.7 on IC-MI).

5. Additional qualitative results

Figure 4 visualizes the learned descriptors used by our method, reduced to three dimensions with PCA, for four representative objects that span textured everyday items (coffee cups, juice boxes) and texture-less industrial parts (gears, metal brackets). Two observations motivate the fused representation. First, within each object the colours are stable across corresponding surface regions, indicating that the descriptors are repeatable and discriminative: exactly the

Table 2. Per-scene effect of refinement across the three datasets. AR (BOP) and ADD-S in % (higher better); RE [°] and TE [mm] (lower better). All runs use identical settings, only the correspondences fed to RANSAC differ (bef.: initial correspondences; aft.: refined). Our refined results are **highlighted**. Imp. denotes the absolute change, computed as aft. — bef. Green indicates improvement or no degradation, while red indicates worse performance.

Dataset	Scene	AR (BOP) [%] ↑			ADD-S [%] ↑			RE [°] ↓			TE [mm] ↓		
		bef.	aft.	Imp.	bef.	aft.	Imp.	bef.	aft.	Imp.	bef.	aft.	Imp.
IC-BIN	000001	37.1	37.5	+0.4	82.2	84.2	+2.0	59.8	59.6	-0.2	15.6	15.3	-0.3
	000002	39.1	39.4	+0.3	83.1	80.2	-2.9	70.7	70.0	-0.7	21.7	24.9	+3.2
IC-MI	000001	64.7	63.0	-1.7	100.0	100.0	0.0	41.6	42.0	+0.4	4.5	4.8	+0.3
	000002	63.4	63.1	-0.3	100.0	100.0	0.0	46.3	46.8	+0.5	2.4	2.6	+0.2
	000003	87.2	84.3	-2.9	100.0	100.0	0.0	9.3	11.2	+1.9	4.1	4.1	0.0
	000004	74.3	74.7	+0.4	100.0	100.0	0.0	32.7	32.7	0.0	3.9	3.6	-0.3
	000005	73.8	72.4	-1.4	100.0	100.0	0.0	15.9	15.8	-0.1	6.0	6.3	+0.3
	000006	72.1	72.1	0.0	100.0	100.0	0.0	23.6	21.7	-1.9	7.3	7.7	+0.4
XYZ-IBD	000000	72.9	75.1	+2.2	96.5	97.1	+0.6	6.8	5.8	-1.0	2.8	2.5	-0.3
	000005	34.8	42.1	+7.3	58.6	66.9	+8.3	51.8	43.2	-8.6	23.0	15.8	-7.2
	000010	61.1	62.3	+1.2	98.3	98.4	+0.1	9.0	8.1	-0.9	4.9	4.7	-0.2
	000015	63.5	65.6	+2.1	93.9	94.7	+0.8	22.0	19.4	-2.6	7.9	7.2	-0.7
	000020	46.0	48.2	+2.2	84.9	86.7	+1.8	24.3	22.3	-2.0	9.0	8.0	-1.0
	000025	61.9	65.7	+3.8	95.6	96.2	+0.6	12.5	6.7	-5.8	6.2	5.4	-0.8
	000030	44.5	47.6	+3.1	87.7	89.8	+2.1	36.8	33.4	-3.4	10.7	8.9	-1.8
	000035	75.3	74.8	-0.5	100.0	100.0	0.0	3.8	3.6	-0.2	6.6	6.4	-0.2
	000040	62.4	67.0	+4.6	98.7	99.2	+0.5	19.3	12.0	-7.3	6.1	5.3	-0.8
	000045	38.9	38.1	-0.8	92.5	92.1	-0.4	69.9	69.9	0.0	10.7	10.5	-0.2
	000050	42.2	43.1	+0.9	93.3	93.7	+0.4	53.6	51.1	-2.5	16.3	15.0	-1.3
	000055	56.0	57.6	+1.6	100.0	100.0	0.0	74.2	69.4	-4.8	32.8	24.5	-8.3
	000060	50.8	57.0	+6.2	88.4	92.1	+3.7	24.1	15.7	-8.4	12.7	10.6	-2.1
	000065	64.6	64.3	-0.3	97.8	98.0	+0.2	28.3	30.2	+1.9	23.0	21.9	-1.1
	000070	30.5	32.1	+1.6	82.0	84.8	+2.8	87.8	85.5	-2.3	13.7	12.6	-1.1

Table 3. Effect of global refinement on relative 6D-pose accuracy, aggregated per dataset. AR (BOP) and ADD-S are reported in % (higher better); RE [°] and TE [mm] are lower better. All runs use identical settings, only the correspondences fed to RANSAC differ (before: initial correspondences; after: refined). Our refined results are **highlighted**. Imp. denotes the absolute change, computed as after — before. Green indicates improvement or no degradation, while red indicates worse performance.

Dataset	AR [%] (BOP) ↑			ADD-S [%] ↑			RE [°] ↓			TE [mm] ↓		
	before	after	Imp.	before	after	Imp.	before	after	Imp.	before	after	Imp.
XYZ-IBD	53.2	55.8	+2.6	90.6	92.1	+1.5	32.13	28.63	-3.50	10.62	9.09	-1.53
IC-BIN	37.6	38.0	+0.4	82.4	83.3	+0.9	62.36	62.03	-0.33	17.01	17.54	+0.53
IC-MI	71.2	70.3	-0.9	100.0	100.0	0.0	30.60	30.74	+0.14	4.49	4.64	+0.15

property our coarse matching relies on. Second, the two modalities are complementary: DINOv2 produces semantic features on textured objects (*e.g.* it separates the label, body, and rim of a cup) but collapses to meaningless representations on the untextured gears and brackets, where appearance carries little information; dGeDi behaves oppositely, encoding local geometry (gear teeth, bracket edges and holes) and thus remaining discriminative precisely where DINOv2 fails, while being less informative on smooth, textured surfaces.

The fused descriptor, obtained by concatenating the two, inherits the strengths of both and stays discriminative across all four categories. This explains why fusion attains the best average recall in our feature ablation (Table 2 of the main paper) without any per-object tuning, and why we adopt it as the default: a single representation that transfers from everyday textured objects to texture-less industrial parts.

Figure 5 shows how the complementary properties observed in the descriptor visualizations translate into the cor-



Figure 4. Qualitative comparison of appearance-aware and semantic-aware, geometry-aware, and fused feature representations for different object categories. From top to bottom: coffee cups, juice boxes, gears, and metal brackets. The left column shows the input object crop, the second column shows DINOv2 features, the third column shows dGeDi features, and the fourth column shows fused features. Here we used Principal Component Analysis (PCA) to reduce the original feature size to 3 for visualization purpose. DINOv2 captures stronger semantic and appearance cues, which are particularly useful for textured objects such as coffee cups and juice boxes. In contrast, dGeDi captures geometric structure more effectively, making it better suited for texture-less industrial objects such as gears and metal brackets.

respondences used for pose estimation. For textured everyday objects, such as coffee cups and juice boxes, DINOv2 produces reliable matches between visually distinctive regions, whereas its correspondences become less discriminative on textureless industrial objects. Conversely, dGeDi yields stronger correspondences on gears and metal brackets by exploiting distinctive geometric structures, such as teeth, edges, and holes, but provides fewer informative matches on

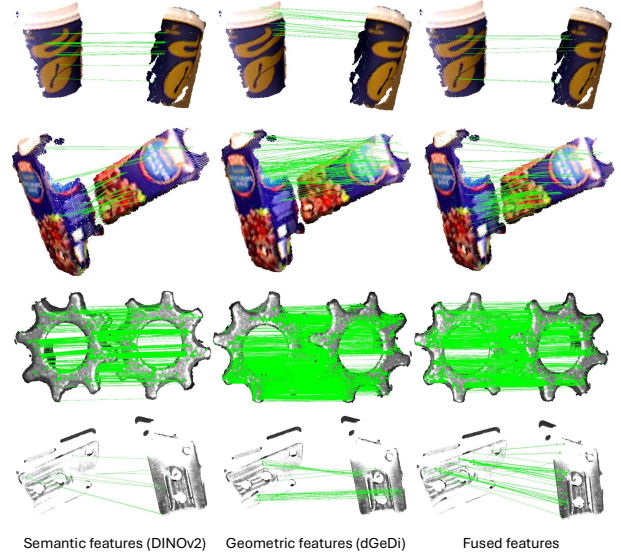


Figure 5. Qualitative comparison of inlier correspondences obtained using semantic (DINOv2), geometric (dGeDi), and fused feature representations. From top to bottom, we show coffee cups, juice boxes, gears, and metal brackets. Semantic features provide stronger appearance cues for textured objects, whereas geometric features are more informative for textureless industrial objects. The fused representation combines these complementary cues and generally produces a denser and more robust set of inlier correspondences across object types.

smooth surfaces. By combining these cues, the fused representation preserves appearance-based matches where texture is informative and introduces geometry-based matches where appearance is ambiguous. Consequently, it generally produces a denser and more spatially distributed set of inlier correspondences across object categories. Together with Figure 4, these results show that fusion improves both descriptor discriminability and its practical outcome during coarse matching, providing a robust representation across the heterogeneous objects considered in our benchmark.

Figure 6 visualizes how the refinement step acts on individual three-instance cycles. Each panel draws, for a triplet (anchor X_k , mediator X_j , target X_i), the two composed correspondences and the direct one; under perfect cycle consistency the composed path and the direct correspondence coincide, so the dashed *closure gap* vanishes. In the success cases (a), the coarse per-pair matches disagree (closure gaps of 42–96 mm and only 0–2 of 3 geometric inliers) and the synchronization resolves them into a fully consistent and correct cycle (3/3 inliers, 0 mm gap). This is the intended behaviour and the source of the accuracy gains reported in the main paper. The failure cases (b) expose the limitation of the approach: refinement always yields a *consistent* cycle (gap $\rightarrow 0$ by construction), but consistency does not imply

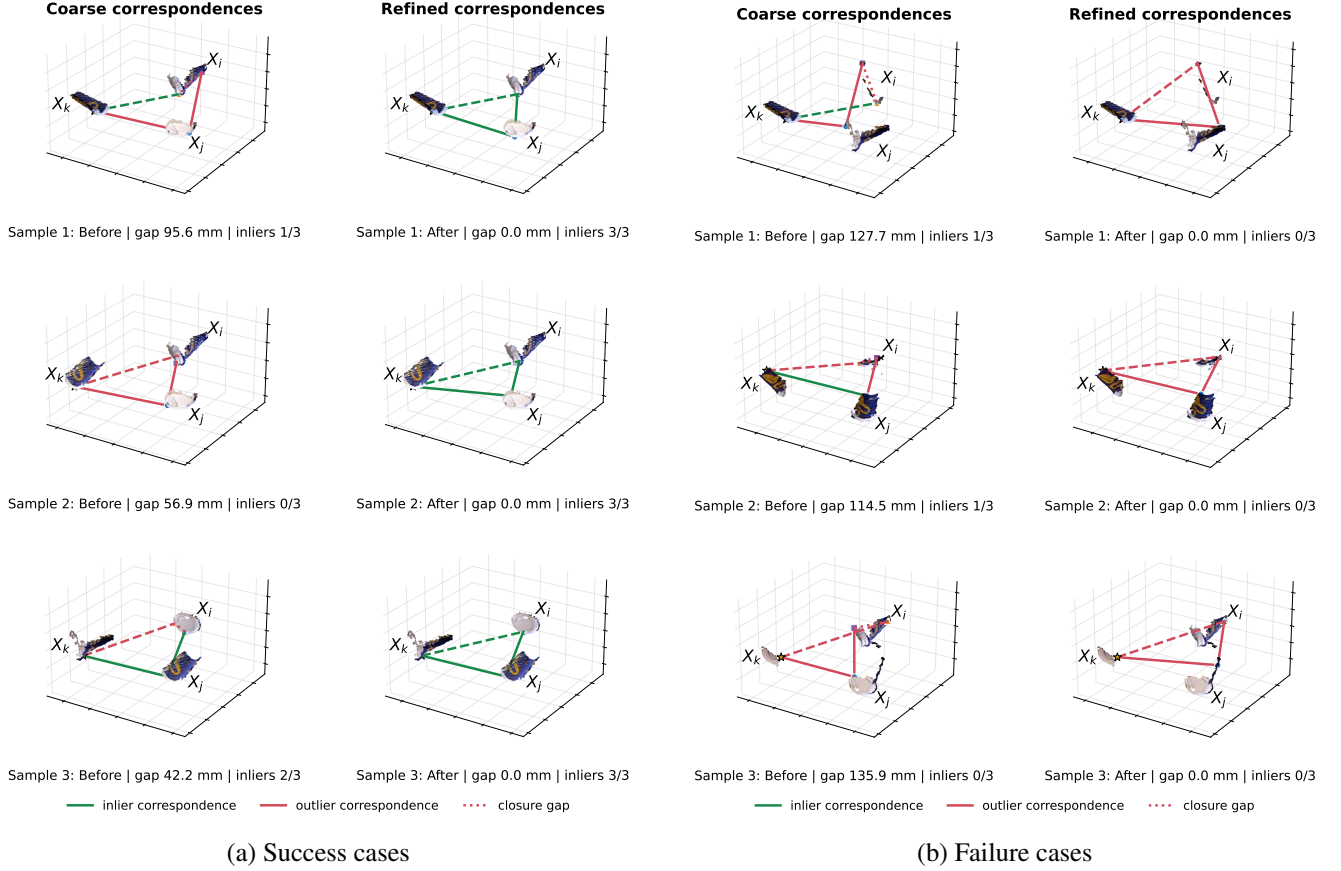


Figure 6. Cycle consistency: successes and failures. Each row is one instance triplet (anchor X_k , mediator X_j , target X_i); the *left* column shows the coarse (independent nearest-neighbour) correspondences and the *right* column the correspondences after cycle-consistency refinement. Green/red lines are symmetry-aware inlier/outlier correspondences and the dashed red line is the cycle-closure gap. (a) Successes: the coarse matches are inconsistent (closure gaps of 42–96 mm and only 0–2 of 3 inliers), and refinement closes the cycle (gap $\rightarrow$ 0) while recovering all three correct edges (3/3 inliers). (b) Failures: refinement still closes the cycle (gap $\rightarrow$ 0) but converges to a geometrically wrong solution (0/3 inliers); in the first two rows it even discards the single correct coarse edge (1/3 $\rightarrow$ 0/3), illustrating that cycle consistency enforces mutual agreement, not ground-truth correctness.

correctness. When the coarse descriptors are ambiguous, *e.g.* repetitive, the low-rank synchronization can lock all three edges onto a mutually agreeing yet wrong assignment (0/3 inliers), and may even overturn a correct coarse edge to reach it (rows 1–2, 1/3 $\rightarrow$ 0/3). Such cases are infrequent in aggregate (cf. Table 2) and are further mitigated downstream by robust RANSAC, but they constitute the principal failure mode of correspondence-level cycle consistency and motivate coupling it with geometric verification.

References

- [1] Andreas Dumanoglou, Rigas Kouskouridas, Sotiris Malasiotis, and Tae-Kyun Kim. Recovering 6D object pose and predicting next-best-view in the crowd. In *CVPR*, 2016. 1
- [2] Amir Hamza, Andrea Caraffa, Davide Boscaini, and Fabio Poiesi. Distilling 3D distinctive local descriptors for 6D pose estimation. In *IROS*, 2025. 1
- [3] Junwen Huang, Jiaqi Hu, Peter KT Yu, Slobodan Ilic, Martin Sundermeyer, and Benjamin Busam. XYZ-IBD: High-precision Bin-picking Dataset for Object 6D Pose Estimation Capturing Real-world Industrial Complexity. *arXiv preprint arXiv:2506.00599*, 2025. 1
- [4] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2024. 1
- [5] Alykhan Tejani, Danhang Tang, Rigas Kouskouridas, and Tae-Kyun Kim. Latent-class hough forests for 3D object detection and pose estimation. In *ECCV*, 2014. 1